

Name _____

Environments and Closures Practice In Class

DAY 1

In Racket, what order are the procedures A B C D E called?

A (1 point)

```
(let ([f (let ([x (A 10)])  
            (lambda ()  
              (B x)))]])  
  
  (C 1)  
  
  (f)  
  
  (f))
```

B (1 point)

```
(let ([x (A 1)]  
      [y (let ([z (B 2)])  
            (lambda () (C z)))]])  
  
  (D 3)  
  
  (y)  
  
  (E 4))
```

C (1 point)

```
((if (equal? (A) 'use-b) B C)  
  
  (D 1)  
  
  (E 2))
```

Draw E&C Diagrams for the following code:

A (4 points)

```
(let ([x 10]
      [y 20])
  (let ([x 5]
        [z (+ x y)])
    (* x z)))
```

B (4 points)

```
(let ([make-adder (lambda (x)
                    (lambda (y) (+ x y)))]])
  (let ([add5 (make-adder 5)])
    (add5 10)))
```

C (5 points)

```
(let ([a 2]
      [b 3])
  (let ([add-a (lambda (x) (+ x a))]
        [mult-b (lambda (x) (* x b))])
    (add-a (mult-b 4))))
```